a) $f(x) = \operatorname{sen} x$

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{\operatorname{sen}(x+\Delta x) - \operatorname{sen} x}{\Delta x} \overset{\text{sen da soma}}{=}$$

$$= \lim_{\Delta x \to 0} \frac{\operatorname{sen} x \cos \Delta x + \operatorname{sen} \Delta x \cos x - \operatorname{sen} x}{\Delta x} =$$

$$= \lim_{\Delta x \to 0} \left[\frac{\operatorname{sen} x (\cos \Delta x - 1) + \operatorname{sen} \Delta x \cos x}{\Delta x} \right] = \lim_{\Delta x \to 0} \frac{\operatorname{sen} x (\cos \Delta x - 1)}{\Delta x} \overset{0}{\overset{(*)}{}} +$$

$$+ \lim_{\Delta x \to 0} \frac{\overset{1^{\circ}\text{ lim. fund.}}{\operatorname{sen} \Delta x} \cos x}{\Delta x} = \cancel{\operatorname{sen} x (0)}^{\,0} + 1 . \cos x = \cos x$$

$$\therefore f(x) = \operatorname{sen} x \Rightarrow f'(x) = \cos x .$$

(*) $$\lim_{\Delta x \to 0} \frac{\cos \Delta x - 1}{\Delta x} = \lim_{\Delta x \to 0} \frac{(\cos \Delta x - 1)}{\Delta x} \cdot \left(\frac{\cos \Delta x + 1}{\cos \Delta x + 1} \right) =$$

$$= \lim_{\Delta x \to 0} \frac{(\cos^2 \Delta x - 1)}{(\cos \Delta x + 1) \Delta x} = \lim_{\Delta x \to 0} \frac{\operatorname{sen}^2 \Delta x}{\Delta x (\cos \Delta x + 1)} =$$

$$= \lim_{\Delta x \to 0} \frac{\overset{1^{\circ}\text{ lim. fund.}}{\operatorname{sen} \Delta x} \cdot \cancel{\operatorname{sen} \Delta x}^{\,0}}{\Delta x \, (\cos \Delta x + 1)} = \frac{0}{2} = 0$$